

```

mysql> use cg_mysql;
Database changed
mysql> /* System informatic functions
/*> It generates the metadata of the system
/*> curdate() = current date
/*> curtime() = current time
/*> now() = current date and time */
mysql> select curdate();
+-----+
| curdate() |
+-----+
| 2026-09-12 |
+-----+
1 row in set (0.05 sec)

mysql> select curtime();
+-----+
| curtime() |
+-----+
| 14:17:15 |
+-----+
1 row in set (0.00 sec)

mysql> select database;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that
corresponds to your MySQL server version for the right syntax to use near '' at
line 1
mysql> select now();
+-----+
| now() |
+-----+
| 2026-09-12 14:17:37 |
+-----+
1 row in set (0.03 sec)

mysql> select database();
+-----+
| database() |
+-----+
| cg_mysql |
+-----+
1 row in set (0.00 sec)

mysql> select version();
+-----+
| version() |
+-----+
| 8.0.36 |
+-----+
1 row in set (0.03 sec)

```

```
mysql> select sysdate();
```

```
+-----+  
| sysdate() |
```

```
+-----+  
| 2026-09-12 14:18:10 |
```

```
+-----+  
1 row in set (0.03 sec)
```

```
mysql> select systime();
```

```
ERROR 1305 (42000): FUNCTION cg_mysql.systime does not exist
```

```
mysql> /* Aggregate functions
```

```
/*> 1. count()
```

```
/*> 2. min()
```

```
/*> 3. max()
```

```
/*> 4. avg()
```

```
/*> 5. sum()
```

```
/*> They are applicable only for numeric datatype */
```

```
mysql> show tables;
```

```
+-----+  
| Tables_in_cg_mysql |
```

```
+-----+  
| backup_emp |
```

```
| company_staff |
```

```
| departments |
```

```
| doctors |
```

```
| emp |
```

```
| employees |
```

```
| orders |
```

```
| products |
```

```
| projects |
```

```
| staff |
```

```
| students |
```

```
| studentscores |
```

```
+-----+  
12 rows in set (0.06 sec)
```

```
mysql> select * from employees;
```

```
+-----+-----+-----+-----+-----+-----+  
| emp_id | emp_name | department | salary | experience | age |
```

```
+-----+-----+-----+-----+-----+-----+  
| 101 | Ravi | IT | 65000.00 | 5 | 28 |
```

```
| 102 | Priya | HR | 45000.00 | 2 | 24 |
```

```
| 103 | Amit | Finance | 75000.00 | 8 | 35 |
```

```
| 104 | Sneha | IT | 55000.00 | 3 | 27 |
```

```
| 105 | Rahul | Sales | 40000.00 | 1 | 22 |
```

```
| 106 | Kiran | Finance | 90000.00 | 12 | 40 |
```

```
| 107 | Anjali | HR | 52000.00 | 4 | 29 |
```

```
| 108 | Suresh | Sales | 68000.00 | 7 | 33 |
```

```
+-----+-----+-----+-----+-----+-----+
```

8 rows in set (0.08 sec)

```
mysql> select count(*) from employees;
```

count(*)
8

1 row in set (0.05 sec)

mysql> -- count function does not count null value, it only counts rows with values

```
mysql> select max(experience) from employees;
```

max(experience)
12

1 row in set (0.06 sec)

```
mysql> select min(experience) as low_experience from employees;
```

low_experience
1

1 row in set (0.00 sec)

```
mysql> select max(salary) as max_salary from employees;
```

max_salary
90000.00

1 row in set (0.04 sec)

```
mysql> select min(salary) as min_salary from employees;
```

min_salary
40000.00

1 row in set (0.00 sec)

```
mysql> select avg(salary) as avg_salary from employees;
```

avg_salary
61250.000000

1 row in set (0.00 sec)

```
mysql> select sum(salary) as total_salary from employees;
```

```
+-----+  
| total_salary |  
+-----+  
|    490000.00 |  
+-----+
```

```
1 row in set (0.00 sec)
```

```
mysql> /* Clauses
```

```
/*> -> In mysql clauses are used to control how data is filtered, grouped,  
sorted or modified in SQL queries.
```

```
/*> -> They make SQL Statements more meaningful and powerful.
```

```
/*>
```

```
/*> where clause: filters records based on specified conditions where clause can  
be used as comparison operators, logical operators and etc. operators.
```

```
/*> syntax: select column_name from table_name where condition;
```

```
/*>
```

```
/*> group by clause: It is used to group rows having the same values in one or  
more columns, it is commonly used with aggregate functions such as min(), max(),  
sum(), count(), avg().
```

```
/*> syntax: select column_name/aggregate_function(column_name) from table_name  
group by column_name;
```

```
/*>
```

```
/*> having clause: It is used to filter grouped records after group by clause.  
It is similar to where but it works on grouped instead of individual rows.
```

```
/*> syntax: select column_name/aggregate_function(column_name) from table_name  
group by column_name having condition;
```

```
/*>
```

```
/*> order by clause: It is used to sort the result set in ascending(ASC) or  
descending(DESC) order
```

```
/*> syntax: select column_name from table_name order by column_name;
```

```
/*> */
```

```
mysql> drop table students;
```

```
Query OK, 0 rows affected (0.09 sec)
```

```
mysql> CREATE TABLE STUDENTS (
```

```
->     StudentID INT AUTO_INCREMENT PRIMARY KEY,
```

```
->     Name VARCHAR(50),
```

```
->     Age INT,
```

```
->     Department VARCHAR(30),
```

```
->     Marks INT,
```

```
->     City VARCHAR(30)
```

```
-> );
```

```
Query OK, 0 rows affected (0.06 sec)
```

```
mysql>
```

```
mysql> select*from students;
```

```
Empty set (0.00 sec)
```

```
mysql> INSERT INTO STUDENTS (Name, Age, Department, Marks, City) VALUES
```

```
-> ('Anil', 20, 'CSE', 85, 'Hyderabad'),  
-> ('Sneha', 22, 'ECE', 75, 'Mumbai'),  
-> ('Ravi', 20, 'CSE', NULL, 'Hyderabad'),  
-> ('Divya', 23, 'MECH', 65, NULL),  
-> ('Kiran', 22, 'ECE', 75, 'Pune'),  
-> ('Varun', 21, 'CSE', 92, 'Chennai'),  
-> ('Priya', 20, 'EEE', 88, 'Delhi'),  
-> ('John', 22, 'MECH', 55, 'Mumbai'),  
-> ('Rekha', 21, 'ECE', 75, 'Pune'),  
-> ('Ramesh', 23, 'EEE', NULL, 'Delhi');
```

```
Query OK, 10 rows affected (0.04 sec)
```

```
Records: 10  Duplicates: 0  Warnings: 0
```

```
mysql>
```

```
mysql> select*from students;
```

StudentID	Name	Age	Department	Marks	City
1	Anil	20	CSE	85	Hyderabad
2	Sneha	22	ECE	75	Mumbai
3	Ravi	20	CSE	NULL	Hyderabad
4	Divya	23	MECH	65	NULL
5	Kiran	22	ECE	75	Pune
6	Varun	21	CSE	92	Chennai
7	Priya	20	EEE	88	Delhi
8	John	22	MECH	55	Mumbai
9	Rekha	21	ECE	75	Pune
10	Ramesh	23	EEE	NULL	Delhi

```
10 rows in set (0.00 sec)
```

```
mysql> select * from students where Marks>80;
```

StudentID	Name	Age	Department	Marks	City
1	Anil	20	CSE	85	Hyderabad
6	Varun	21	CSE	92	Chennai
7	Priya	20	EEE	88	Delhi

```
3 rows in set (0.03 sec)
```

```
mysql> select * from students where Age>22;
```

StudentID	Name	Age	Department	Marks	City
4	Divya	23	MECH	65	NULL
10	Ramesh	23	EEE	NULL	Delhi

```
2 rows in set (0.00 sec)
```

```
mysql> select * from students where Department>'CSE';
```

StudentID	Name	Age	Department	Marks	City
2	Sneha	22	ECE	75	Mumbai
4	Divya	23	MECH	65	NULL
5	Kiran	22	ECE	75	Pune
7	Priya	20	EEE	88	Delhi
8	John	22	MECH	55	Mumbai
9	Rekha	21	ECE	75	Pune
10	Ramesh	23	EEE	NULL	Delhi

7 rows in set (0.00 sec)

```
mysql> select * from students where Department='CSE';
```

StudentID	Name	Age	Department	Marks	City
1	Anil	20	CSE	85	Hyderabad
3	Ravi	20	CSE	NULL	Hyderabad
6	Varun	21	CSE	92	Chennai

3 rows in set (0.00 sec)

```
mysql> select * from students where City='Delhi';
```

StudentID	Name	Age	Department	Marks	City
7	Priya	20	EEE	88	Delhi
10	Ramesh	23	EEE	NULL	Delhi

2 rows in set (0.00 sec)

```
mysql> select * from students where department='CSE' and marks>90;
```

StudentID	Name	Age	Department	Marks	City
6	Varun	21	CSE	92	Chennai

1 row in set (0.00 sec)

```
mysql> select * from students where department='CSE' having marks>90;
```

StudentID	Name	Age	Department	Marks	City
6	Varun	21	CSE	92	Chennai

1 row in set (0.00 sec)

```
mysql> -- Group By
```

```
mysql> select department, count(*) from students group by department;
```

department	count(*)
CSE	3
ECE	3
MECH	2
EEE	2

```
4 rows in set (0.04 sec)
```

```
mysql> select marks, average(*) from students group by department;
```

ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near '*) from students group by department' at line 1

```
mysql> select department, average(marks) from students group by department;
```

ERROR 1305 (42000): FUNCTION cg_mysql.average does not exist

```
mysql> select department, avg(marks) from students group by department;
```

department	avg(marks)
CSE	88.5000
ECE	75.0000
MECH	60.0000
EEE	88.0000

```
4 rows in set (0.04 sec)
```

```
mysql> select department, max(marks) from students group by department;
```

department	max(marks)
CSE	92
ECE	75
MECH	65
EEE	88

```
4 rows in set (0.04 sec)
```

```
mysql> select department, min(marks) from students group by department;
```

department	min(marks)
CSE	85
ECE	75
MECH	55
EEE	88

```
4 rows in set (0.00 sec)
```

```
mysql> select department, sum(marks) from students group by department;
```

department	sum(marks)
CSE	177
ECE	225
MECH	120
EEE	88

```
4 rows in set (0.00 sec)
```

```
mysql> select city,count(*) from students group by city;
```

city	count(*)
Hyderabad	2
Mumbai	2
NULL	1
Pune	2
Chennai	1
Delhi	2

```
6 rows in set (0.00 sec)
```

```
mysql> select age,count(*) from students group by age;
```

age	count(*)
20	3
22	3
23	2
21	2

```
4 rows in set (0.00 sec)
```

```
mysql> select department, count(*) as total_students from students group by department having total_students>2;
```

department	total_students
CSE	3
ECE	3

```
2 rows in set (0.00 sec)
```

```
mysql> select department, max(marks) as highest_marks from students group by department having highest_marks>80;
```

department	highest_marks
------------	---------------

CSE	92
EEE	88

2 rows in set (0.00 sec)

```
mysql> select department, avg(marks) as average_marks from students group by
department having average_marks>75;
```

department	average_marks
CSE	88.5000
EEE	88.0000

2 rows in set (0.00 sec)

```
mysql> select city, count(*) as number_of_students from students group by city
having number_of_students>1;
```

city	number_of_students
Hyderabad	2
Mumbai	2
Pune	2
Delhi	2

4 rows in set (0.00 sec)

```
mysql> select department, sum(marks) as average_marks from students group by
department having average_marks>150;
```

department	average_marks
CSE	177
ECE	225

2 rows in set (0.00 sec)

```
mysql> select city, avg(marks) as avg_marks from students group by city having
avg_marks>70;
```

city	avg_marks
Hyderabad	85.0000
Pune	75.0000
Chennai	92.0000
Delhi	88.0000

4 rows in set (0.00 sec)

```
mysql> select department, min(marks) as lowest_marks from students group by
department having lowest_marks<60;
```

```
+-----+-----+
| department | lowest_marks |
+-----+-----+
| MECH       |          55 |
+-----+-----+
1 row in set (0.00 sec)
```

```
mysql> select department, max(marks) as highest_marks from students group by
department having highest_marks>90;
```

```
+-----+-----+
| department | highest_marks |
+-----+-----+
| CSE        |          92 |
+-----+-----+
1 row in set (0.00 sec)
```

```
mysql> select city, sum(marks) as total_marks from students group by city having
total_marks>100;
```

```
+-----+-----+
| city      | total_marks |
+-----+-----+
| Mumbai   |         130 |
| Pune     |         150 |
+-----+-----+
2 rows in set (0.00 sec)
```

```
mysql> -- order by clause
mysql> select * from students;
```

```
+-----+-----+-----+-----+-----+-----+
| StudentID | Name   | Age | Department | Marks | City      |
+-----+-----+-----+-----+-----+-----+
|          1 | Anil   | 20  | CSE        | 85    | Hyderabad |
|          2 | Sneha  | 22  | ECE        | 75    | Mumbai    |
|          3 | Ravi   | 20  | CSE        | NULL  | Hyderabad |
|          4 | Divya  | 23  | MECH       | 65    | NULL      |
|          5 | Kiran  | 22  | ECE        | 75    | Pune      |
|          6 | Varun  | 21  | CSE        | 92    | Chennai   |
|          7 | Priya  | 20  | EEE        | 88    | Delhi     |
|          8 | John   | 22  | MECH       | 55    | Mumbai    |
|          9 | Rekha  | 21  | ECE        | 75    | Pune      |
|         10 | Ramesh | 23  | EEE        | NULL  | Delhi     |
+-----+-----+-----+-----+-----+-----+
10 rows in set (0.00 sec)
```

```
mysql> select * from students order by name;
```

```
+-----+-----+-----+-----+-----+-----+
| StudentID | Name   | Age | Department | Marks | City      |
+-----+-----+-----+-----+-----+-----+
```

1	Anil	20	CSE	85	Hyderabad
4	Divya	23	MECH	65	NULL
8	John	22	MECH	55	Mumbai
5	Kiran	22	ECE	75	Pune
7	Priya	20	EEE	88	Delhi
10	Ramesh	23	EEE	NULL	Delhi
3	Ravi	20	CSE	NULL	Hyderabad
9	Rekha	21	ECE	75	Pune
2	Sneha	22	ECE	75	Mumbai
6	Varun	21	CSE	92	Chennai

10 rows in set (0.05 sec)

```
mysql> select * from students order by name desc;
```

StudentID	Name	Age	Department	Marks	City
6	Varun	21	CSE	92	Chennai
2	Sneha	22	ECE	75	Mumbai
9	Rekha	21	ECE	75	Pune
3	Ravi	20	CSE	NULL	Hyderabad
10	Ramesh	23	EEE	NULL	Delhi
7	Priya	20	EEE	88	Delhi
5	Kiran	22	ECE	75	Pune
8	John	22	MECH	55	Mumbai
4	Divya	23	MECH	65	NULL
1	Anil	20	CSE	85	Hyderabad

10 rows in set (0.00 sec)

```
mysql> select * from students order by city;
```

StudentID	Name	Age	Department	Marks	City
4	Divya	23	MECH	65	NULL
6	Varun	21	CSE	92	Chennai
7	Priya	20	EEE	88	Delhi
10	Ramesh	23	EEE	NULL	Delhi
1	Anil	20	CSE	85	Hyderabad
3	Ravi	20	CSE	NULL	Hyderabad
2	Sneha	22	ECE	75	Mumbai
8	John	22	MECH	55	Mumbai
5	Kiran	22	ECE	75	Pune
9	Rekha	21	ECE	75	Pune

10 rows in set (0.00 sec)

```
mysql> select * from students order by age desc;
```

StudentID	Name	Age	Department	Marks	City
-----------	------	-----	------------	-------	------

4	Divya	23	MECH	65	NULL
10	Ramesh	23	EEE	NULL	Delhi
2	Sneha	22	ECE	75	Mumbai
5	Kiran	22	ECE	75	Pune
8	John	22	MECH	55	Mumbai
6	Varun	21	CSE	92	Chennai
9	Rekha	21	ECE	75	Pune
1	Anil	20	CSE	85	Hyderabad
3	Ravi	20	CSE	NULL	Hyderabad
7	Priya	20	EEE	88	Delhi

10 rows in set (0.00 sec)

mysql> /* Executon order of clauses

/*> 1. from

/*> 2. where

/*> 3. group by

/*> 4. having

/*> 5. select

/*> 6. order by

/*> */

mysql> select department, avg(marks) as average_marks from students where age> 20
having average_marks > 70 order by average_marks desc;

ERROR 1140 (42000): In aggregated query without GROUP BY, expression #1 of SELECT
list contains nonaggregated column 'cg_mysql.students.Department'; this is
incompatible with sql_mode=only_full_group_by

mysql> select department, avg(marks) as average_marks from students where age> 20
group by department having average_marks > 70 order by average_marks desc;

department	average_marks
CSE	92.0000
ECE	75.0000

2 rows in set (0.00 sec)

mysql> select city, count(*) as total_students from students where marks>70 group
by city having count(*)>1 order by total_students;

city	total_students
Pune	2

1 row in set (0.00 sec)

mysql> select department, sum(marks) as total_marks from students where students in
('Hyderabad', 'Pune') group by department having total_marks > 100 order by
total_marks;

ERROR 1054 (42S22): Unknown column 'students' in 'where clause'

```
mysql> select department, sum(marks) as total_marks from students where city in ('Hyderabad','Pune') group by department having total_marks > 100 order by total_marks;
```

```
+-----+-----+
| department | total_marks |
+-----+-----+
| ECE        |          150 |
+-----+-----+
1 row in set (0.04 sec)
```

```
mysql> select city, max(marks) as highest_marks from students where department = 'ECE' group by city having highest_marks>70 order by highest_marks;
```

```
+-----+-----+
| city   | highest_marks |
+-----+-----+
| Mumbai |          75   |
| Pune   |          75   |
+-----+-----+
2 rows in set (0.00 sec)
```

```
mysql> select department, avg(age) as average_age from students where city is not null group by department having average_age>90 order by name;
ERROR 1055 (42000): Expression #1 of ORDER BY clause is not in GROUP BY clause and contains nonaggregated column 'cg_mysql.students.Name' which is not functionally dependent on columns in GROUP BY clause; this is incompatible with sql_mode=only_full_group_by
mysql> select department, avg(age) as average_age from students where city is not null group by department having average_age>90 order by department;
Empty set (0.00 sec)
```

```
mysql> CREATE TABLE sales (
->     sale_id INT PRIMARY KEY,
->     product VARCHAR(50),
->     category VARCHAR(50),
->     units_sold INT,
->     unit_price INT,
->     region VARCHAR(50)
-> );
Query OK, 0 rows affected (0.05 sec)
```

```
mysql>
mysql> INSERT INTO sales (sale_id, product, category, units_sold, unit_price, region) VALUES
-> (1, 'Keyboard', 'Electronics', 10, 1200, 'North'),
-> (2, 'Monitor', 'Electronics', 5, 7000, 'South'),
-> (3, 'Chair', 'Furniture', 15, 2500, 'North'),
-> (4, 'Desk', 'Furniture', 7, 4500, 'West'),
-> (5, 'Mouse', 'Electronics', 20, 800, 'East'),
-> (6, 'Sofa', 'Furniture', 3, 15000, 'South'),
-> (7, 'Headphones', 'Electronics', 8, 1800, 'North'),
```

```

-> (8, 'Laptop', 'Electronics', 6, 55000, 'West'),
-> (9, 'Table', 'Furniture', 12, 6000, 'East'),
-> (10, 'Fan', 'HomeAppliance', 18, 3000, 'South'),
-> (11, 'AC', 'HomeAppliance', 4, 35000, 'North'),
-> (12, 'Cupboard', 'Furniture', 5, 12000, 'West'),
-> (13, 'Printer', 'Electronics', 9, 9000, 'East'),
-> (14, 'Bed', 'Furniture', 2, 25000, 'South'),
-> (15, 'Mobile', 'Electronics', 14, 20000, 'North');

```

Query OK, 15 rows affected (0.04 sec)

Records: 15 Duplicates: 0 Warnings: 0

```
mysql> select * from sales;
```

sale_id	product	category	units_sold	unit_price	region
1	Keyboard	Electronics	10	1200	North
2	Monitor	Electronics	5	7000	South
3	Chair	Furniture	15	2500	North
4	Desk	Furniture	7	4500	West
5	Mouse	Electronics	20	800	East
6	Sofa	Furniture	3	15000	South
7	Headphones	Electronics	8	1800	North
8	Laptop	Electronics	6	55000	West
9	Table	Furniture	12	6000	East
10	Fan	HomeAppliance	18	3000	South
11	AC	HomeAppliance	4	35000	North
12	Cupboard	Furniture	5	12000	West
13	Printer	Electronics	9	9000	East
14	Bed	Furniture	2	25000	South
15	Mobile	Electronics	14	20000	North

15 rows in set (0.00 sec)

```
mysql> notee
```